

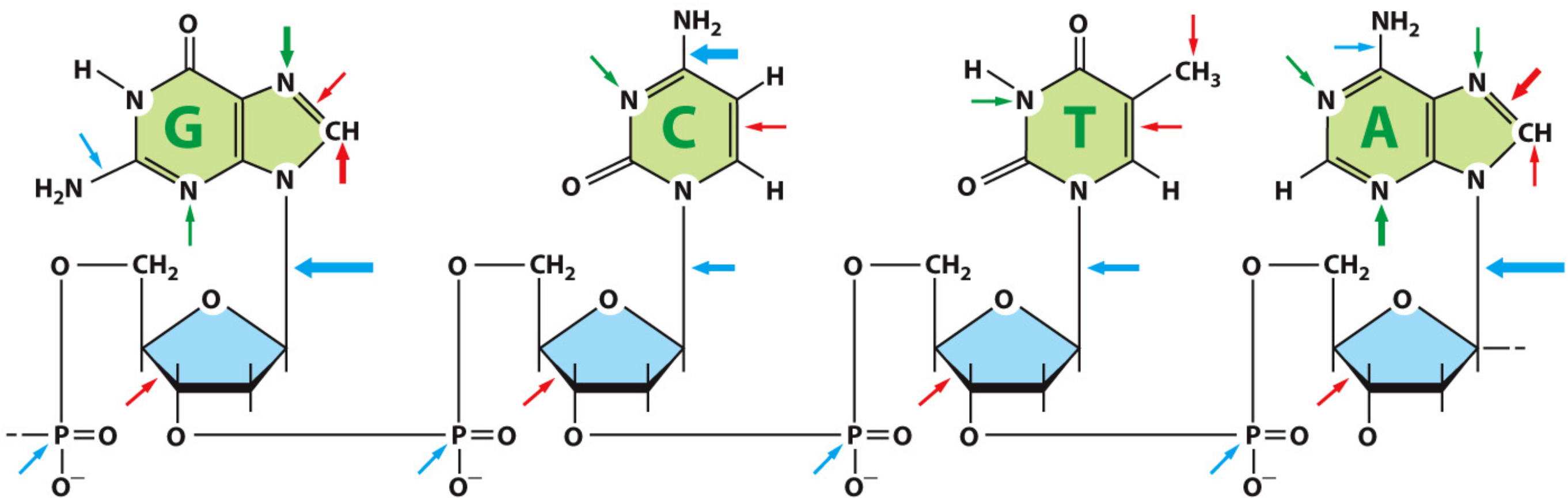
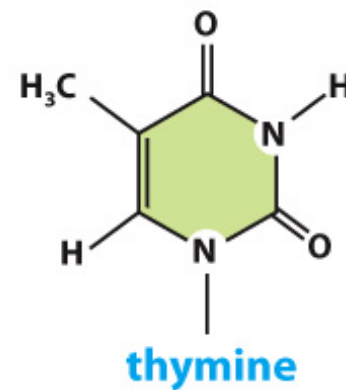
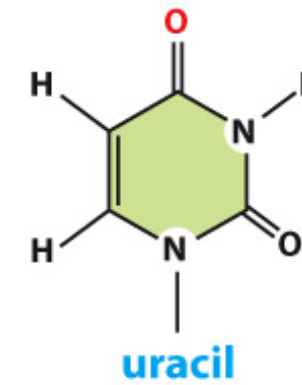
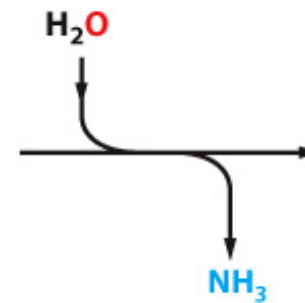
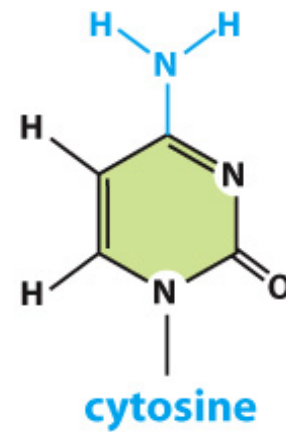
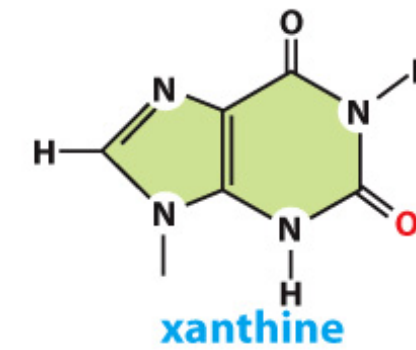
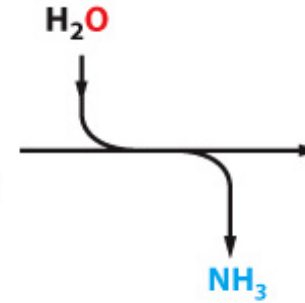
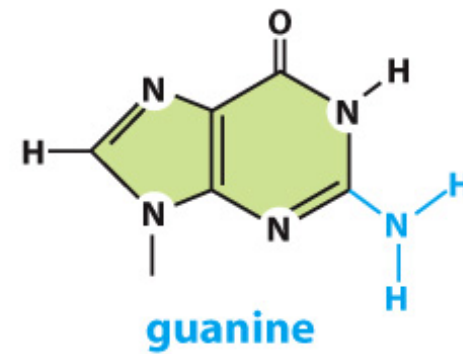
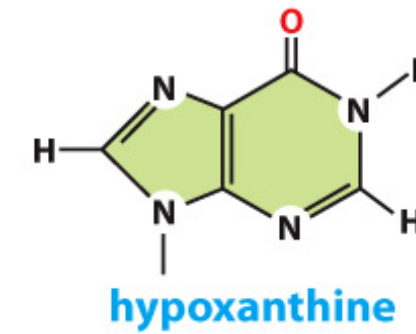
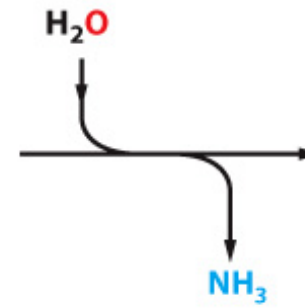
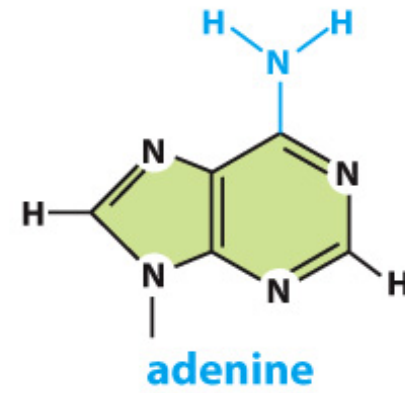


Figure 5-37 Molecular Biology of the Cell 6e (© Garland Science 2015)

NATURAL DNA BASES

UNNATURAL DNA BASES



NO DEAMINATION

Figure 5-43a Molecular Biology of the Cell 6e (© Garland Science 2015)

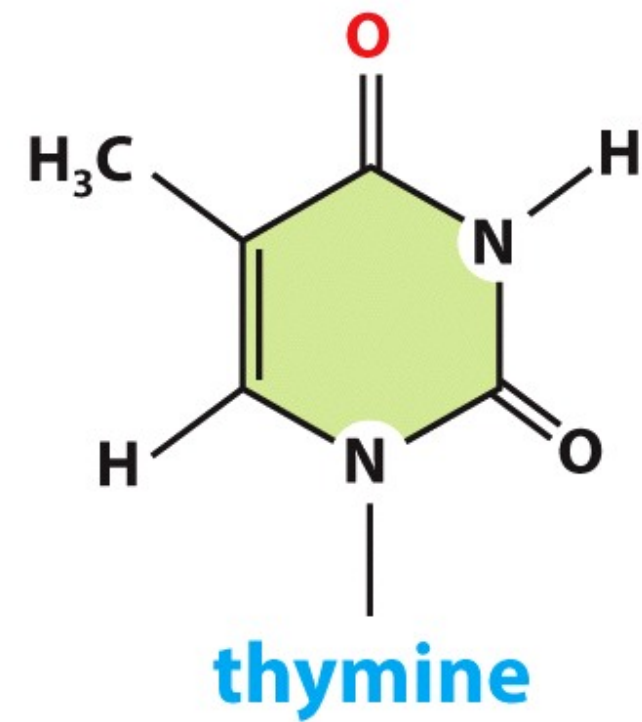
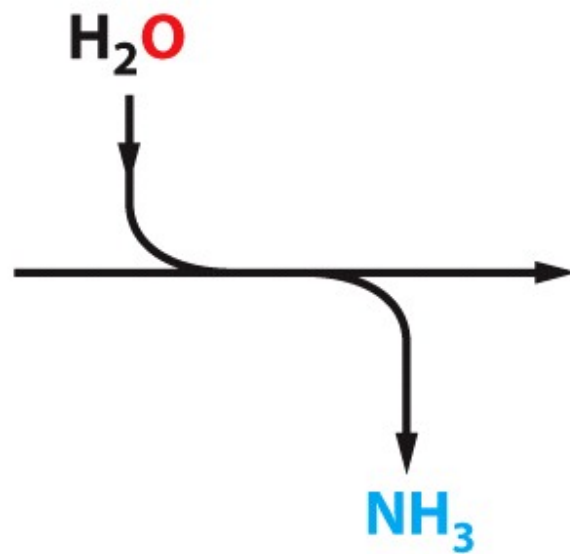
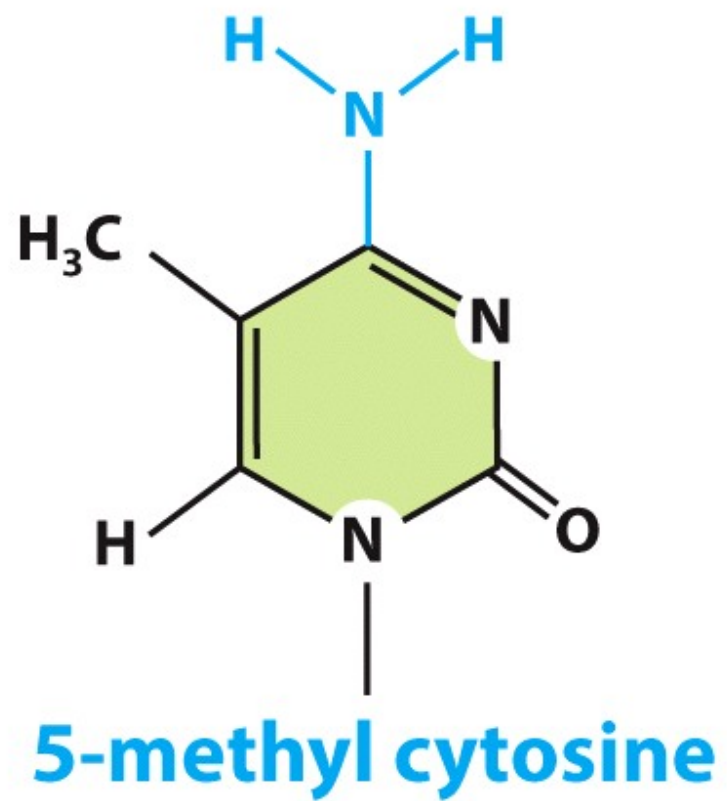


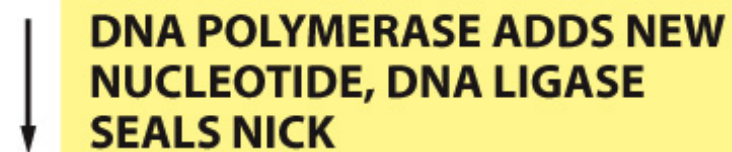
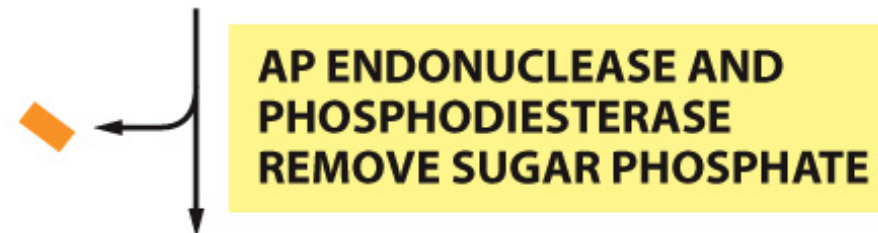
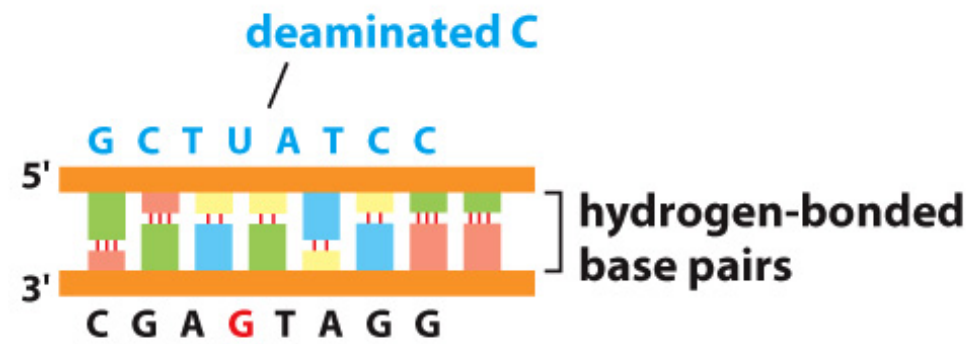
Figure 5-43b Molecular Biology of the Cell 6e (© Garland Science 2015)

TABLE 5–3 Endogenous DNA Lesions Arising and Repaired in a Diploid Mammalian Cell in 24 Hours

DNA lesion	Number repaired in 24 h
Hydrolysis	
Depurination	18,000
Depyrimidination	600
Cytosine deamination	100
5-Methylcytosine deamination	10
Oxidation	
8-oxo G	1500
Ring-saturated pyrimidines (thymine glycol, cytosine hydrates)	2000
Lipid peroxidation products (M1G, etheno-A, etheno-C)	1000
Nonenzymatic methylation by S-adenosylmethionine	
7-Methylguanine	6000
3-Methyladenine	1200
Nonenzymatic methylation by nitrosated polyamines and peptides	
O ⁶ -Methylguanine	20–100

The DNA lesions listed in the table are the result of the normal chemical reactions that take place in cells. Cells that are exposed to external chemicals and radiation suffer greater and more diverse forms of DNA damage. (From T. Lindahl and D.E. Barnes, *Cold Spring Harb. Symp. Quant. Biol.* 65:127–133, 2000.)

BASE EXCISION REPAIR



NUCLEOTIDE EXCISION REPAIR

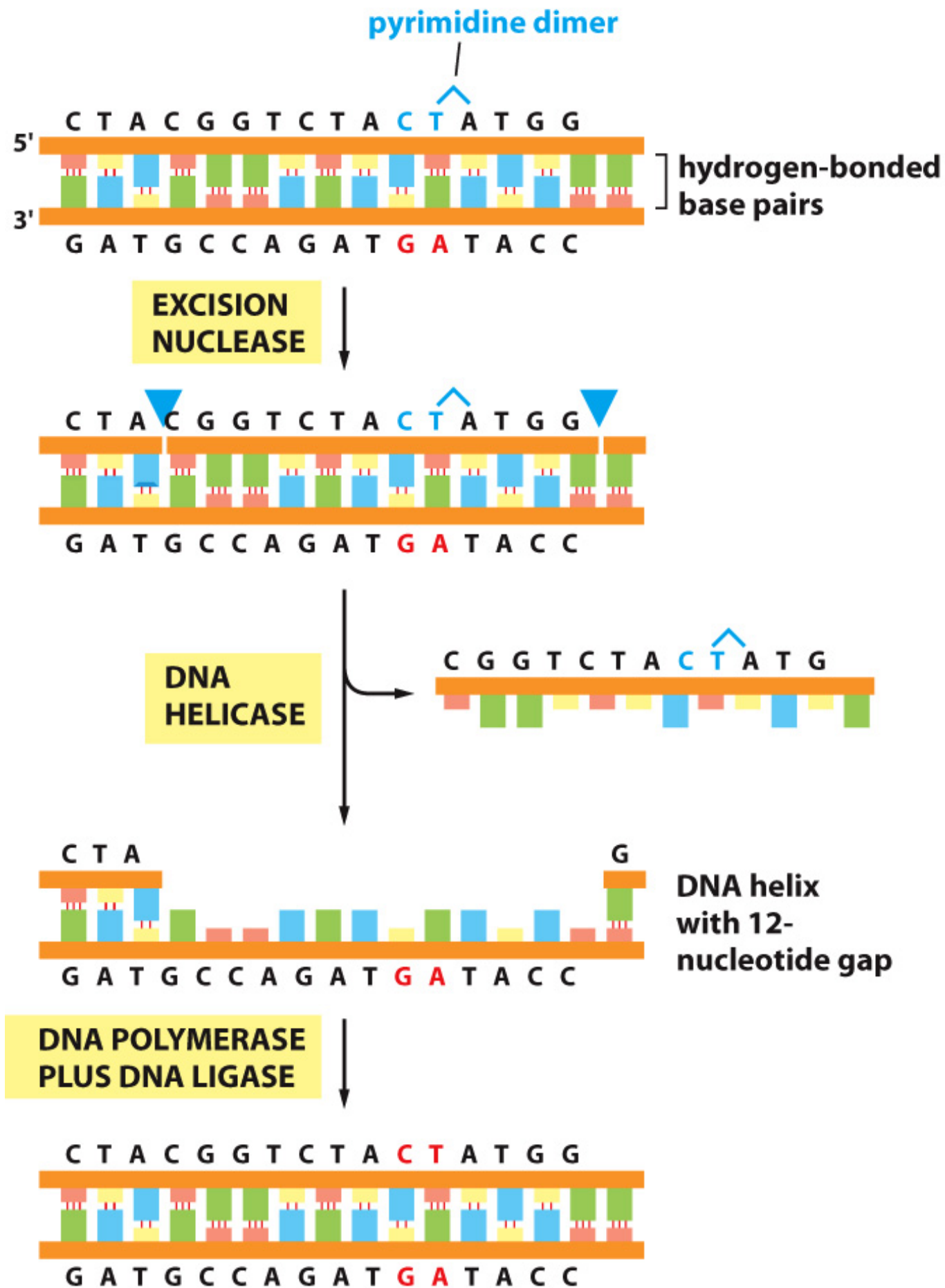


Figure 5-41b Molecular Biology of the Cell 6e (© Garland Science 2015)

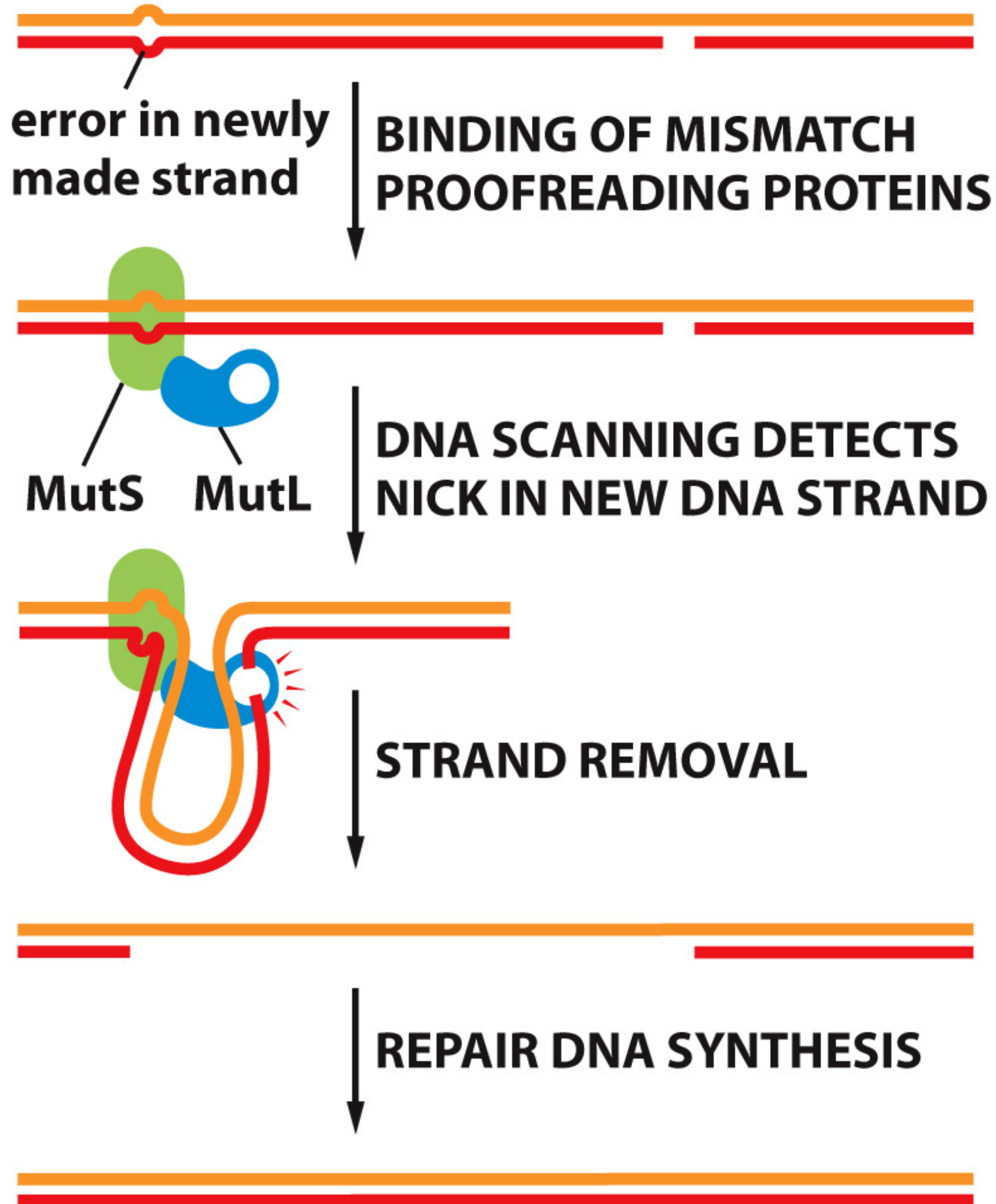


Figure 5-19a Molecular Biology of the Cell 6e (© Garland Science 2015)

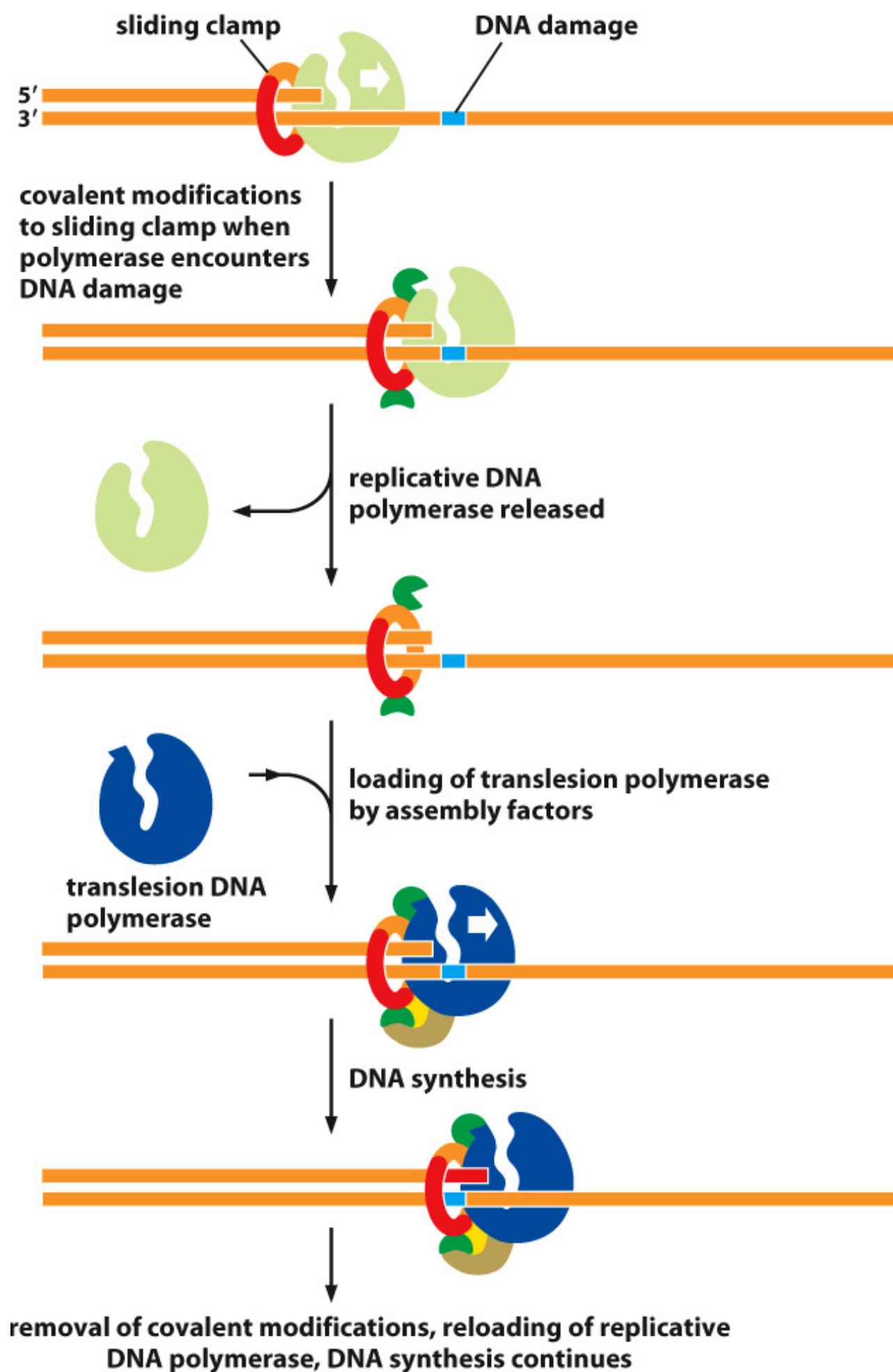
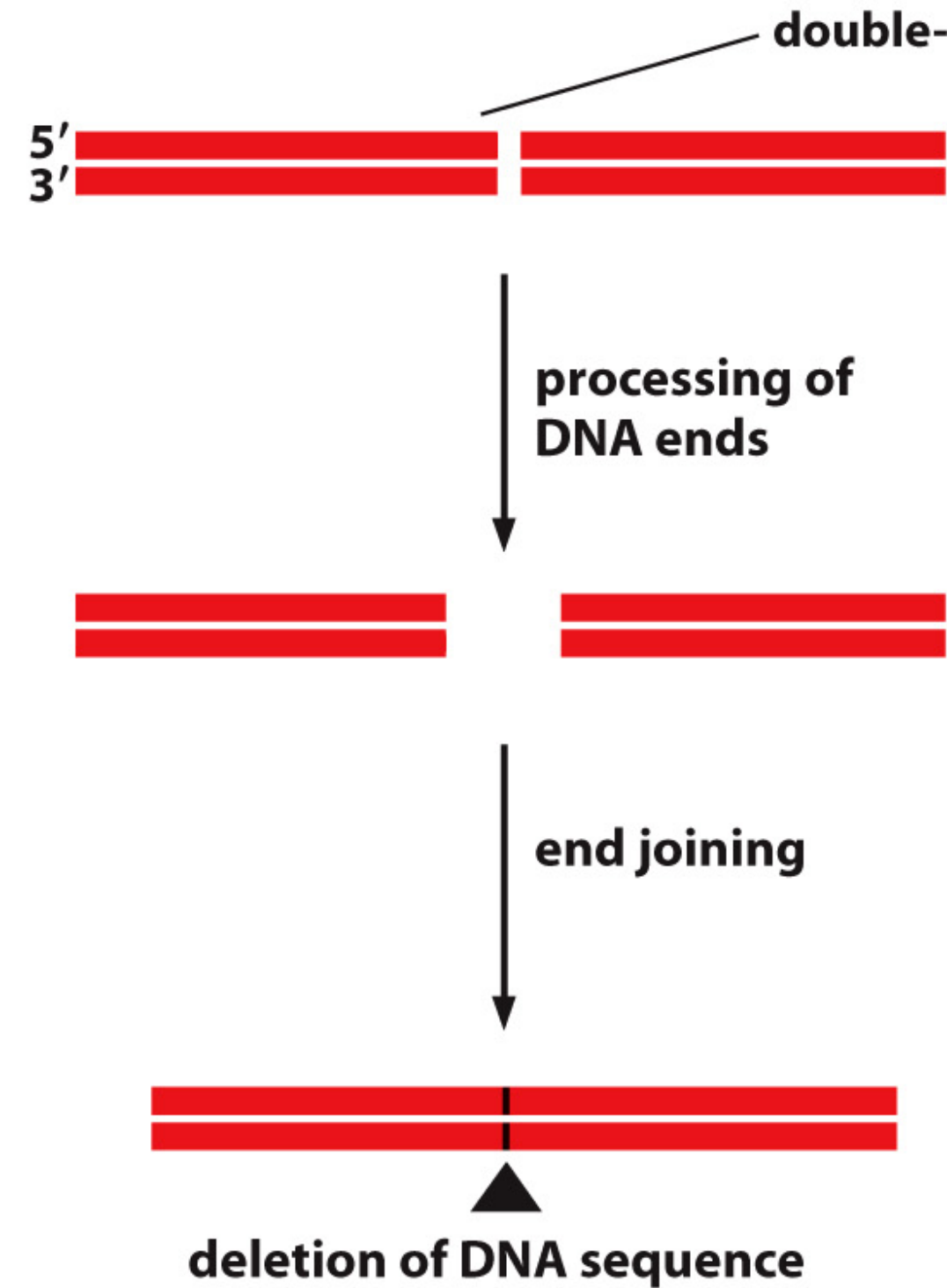


Figure 5-44 Molecular Biology of the Cell 6e (© Garland Science 2015)

(A) NONHOMOLOGOUS END JOINING



(B) HOMOLOGOUS RECOMBINATION

